Response-Curvature Rigidity for Chiral Shear Transport: An Exact Circle Classification and a Positive-Dissipation Counterexample

Alex Moraru

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Abstract

We isolate a geometric rigidity criterion for a two-component chiral transport response. Let the shifted complex viscosity be $Z(p) = \eta_s(p) + i\eta_{\mathrm{H}}(p)$ and let the achiral response be $Z_0 = x_0 > 0$. The circle having 0 and x_0 as endpoints of a diameter is obeyed if and only if the real part of the actual inverse response is chirality independent, $\mathrm{Re}[1/Z(p)] = \mathrm{Re}[1/Z_0]$. This statement is exact and does not require molecular chaos, a fixed susceptibility, or a single-relaxation-time closure. For a C_6-covariant two-dimensional shear block, the commutant is $aI + bJ$; C_6 provides this two-coordinate form but does not impose the required rigidity of a. We derive the corresponding response-curvature formula, reproduce the molecular-chaos chiral-FHP complex shear eigenvalue by exhaustive enumeration of all 64 local Boolean states, and verify the published shear/Hall circle exactly. We also construct a stationary Ornstein–Uhlenbeck Green–Kubo model with strict positive dissipation and the same reduced rotational/reflection covariance for which a drifts and the circle fails. The result is therefore a classification with explicit witnesses on both sides, not a proof that the fully correlated FHP automaton belongs to the rigid class.

1 Response-level circle theorem

Write

$$Z = x + iy \neq 0, \qquad Z_0 = x_0 > 0. \tag{1}$$

The anchored circle is

$$(x - x_0/2)^2 + y^2 = (x_0/2)^2, \tag{2}$$

which is equivalent to $\Delta = x^2 + y^2 - x_0 x = 0$. Introduce the actual inverse response

$$Q = \frac{1}{Z} = a + ib, \qquad a = \frac{x}{x^2 + y^2}. \tag{3}$$

Then

$$\frac{\Delta}{|Z|^2} = 1 - x_0 \,\mathrm{Re}\,\frac{1}{Z}. \tag{4}$$

At the achiral point $a_0 = 1/x_0$. Hence Eq. (2) holds pointwise if and only if

$$\mathrm{Re}\,\frac{1}{Z(p)} = a_0. \tag{5}$$

This is the central rigidity criterion. It is a statement about the measured response itself, so any chirality-dependent residue or susceptibility that may occur in a kinetic representation is already included.

For $Q = a + ib$, inversion yields the exact defect

$$\Delta = \frac{a_0 - a}{a_0(a^2 + b^2)}. \tag{6}$$

Reactive changes of b therefore move a point along the anchored circle when $a = a_0$, while drift of the real inverse response moves the point off that circle.

2 C_6 shear commutant

The traceless-symmetric shear tensor in two spatial dimensions has two independent components. Under a physical rotation through angle θ, this doublet rotates by 2θ. Thus a generator of C_6 acts as a nontrivial $2\pi/3$ rotation R on the shear doublet. Solving $XR = RX$ for a real 2×2 matrix gives

$$X = aI + bJ, \qquad J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad J^2 = -I. \tag{7}$$

Reflection conjugates J to $-J$. With $d = p - p_0$, reflection covariance consequently permits $a(-d) = a(d)$ and $b(-d) = -b(d)$. It does not require $a(d)$ to be constant. The rigidity condition in Eq. (5) is therefore additional to C_6 symmetry.

3 Response curvature

Let $a(d)$ be even, $b(d)$ odd, $a(0) = a_0 > 0$, $a''(0) = a_2$, and $b'(0) = b_1 > 0$. For the planar response curve $Z(d) = 1/[a(d) + ib(d)]$, direct differentiation gives the signed achiral curvature

$$\kappa(0) = -2a_0 - \frac{a_0^2 a_2}{b_1^2}. \tag{8}$$

For the rigid class $a = a_0$, inversion of the vertical line $\mathrm{Re}\, Q = a_0$ is the circle in Eq. (2), with radius $1/(2a_0)$ and signed curvature $-2a_0$ in the chosen orientation.

Equation (8) detects leading quadratic dissipative drift. It does not make a single local curvature value a global classifier: $a(d) = a_0 + \epsilon d^4$ has $a''(0) = 0$ and therefore the same achiral curvature as the rigid class, while Eq. (6) is nonzero at every $d \neq 0$. The complete global criterion is Eq. (5).

4 Molecular-chaos FHP witness

Allocca et al. derive for their zero-field chiral FHP automaton a complex shear collision eigenvalue

$$\lambda = -\frac{\rho}{2}\left(1 - \frac{\rho}{6}\right)^3 \left[1 - i\frac{2\sqrt{3}}{3}\left(p - \frac{1}{2}\right)\right] \tag{9}$$

and the kinematic viscosities

$$\eta = \frac{1}{2\rho(1-\rho/6)^3}\frac{1}{1 + \frac{4}{3}(p - \frac{1}{2})^2} - \frac{1}{8}, \tag{10}$$

$$\eta_{\mathrm{H}} = -\frac{\sqrt{3}}{3\rho(1-\rho/6)^3}\frac{p - \frac{1}{2}}{1 + \frac{4}{3}(p - \frac{1}{2})^2}. \tag{11}$$

The repository independently enumerates all 64 six-bit local collision states, constructs the Bernoulli molecular-chaos collision map, differentiates it at homogeneous filling $n_\ell = \rho/6$, and Fourier-diagonalizes the resulting circulant Jacobian. The shear eigenvalue agrees symbolically with Eq. (9).

Writing $D = \rho(1-\rho/6)^3$, $A = 1/(2D)$ and $u = (2\sqrt{3}/3)(p-1/2)$ gives

$$Z_B = \eta + \frac{1}{8} + i\eta_{\mathrm{H}} = \frac{A}{1+iu} = -\frac{1}{4\lambda^*}. \tag{12}$$

Hence the molecular-chaos response lies identically on the anchored circle. In addition to the symbolic proof, the code checks 9,595 exact rational grid points.

5 Positive-dissipation Green–Kubo counterexample

Consider the stationary Ornstein–Uhlenbeck process

$$dX = -K(d)X\,dt + B(d)\,dW, \tag{13}$$

with

$$K(d) = [a_0 + \epsilon d^2]I + cdJ, \qquad a_0, \epsilon, c > 0. \tag{14}$$

Choose stationary covariance $C = AI$. The Lyapunov equation is satisfied by

$$BB^T = KC + CK^T = 2A[a_0 + \epsilon d^2]I > 0. \tag{15}$$

The symmetric part of K is strictly positive, the eigenvalues have positive real part, and the process is stable. The exact correlation integral is

$$\int_0^\infty e^{-Kt} C\,dt = K^{-1}C. \tag{16}$$

After dividing the known scalar covariance factor, the response is $Z = 1/(a+ib)$ and

$$\Delta(d) = -\frac{\epsilon d^2}{a_0\{[a_0+\epsilon d^2]^2 + c^2 d^2\}}, \tag{17}$$

which is strictly negative for $d \neq 0$. Thus C_6/reflection covariance and positive dissipation do not protect the circle.

6 Spectral criterion and its limit

For an already isolated reduced shear block of the form Eq. (7), the eigenvalues are $a \pm ib$. Once the response residue/normalization relating that block to $Q = 1/Z$ is known, Eq. (5) is decidable algebraically from the reduced spectrum without time-domain simulation.

This does not extend to a bare unordered spectrum of a larger generator. Resolvent matrix elements also depend on spectral projectors and observable overlaps. The code contains two similar matrices with identical eigenvalues but different $g^T L^{-1} g$ for a fixed g, providing an exact obstruction to the over-broad spectral claim.

7 Renormalization and Ward-identity status

At scale ℓ, define $Q_\ell = a_\ell + ib_\ell$. The rigid manifold is

$$a_\ell(d) - a_\ell(0) = 0. \tag{18}$$

A scale flow $\partial_\ell a = \beta_a$ preserves this manifold if and only if it is tangent:

$$\beta_a(d) - \beta_a(0) = 0 \tag{19}$$

when the rigidity constraint holds. C_6 alone does not imply this relation.

The exact defect formula resembles a response constraint but is not, by itself, a Ward identity. A genuine Ward interpretation requires an identified symmetry or conservation law that enforces the tangency condition. Established Kubo/Ward literature supplies examples of such symmetry-generated identities, but no such microscopic protection has yet been derived here for the correlated chiral FHP automaton.

8 Two-dimensional Green–Kubo qualification

Long-time tails in two-dimensional FHP fluids make an unregulated thermodynamic-limit zero-frequency shear viscosity delicate. The structural criterion can instead be applied at fixed finite size, frequency or renormalization scale:

$$\mathrm{Re}[1/Z_{L,\omega}(p)] = \mathrm{Re}[1/Z_{L,\omega}(p_0)]. \tag{20}$$

Whether correlated chiral FHP satisfies this condition is therefore an open microscopic question. A model-specific proof of CRTC survival must establish the condition under the chosen limiting procedure; it cannot be inferred from the molecular-chaos circle alone.

9 Reproducibility

The accompanying repository contains exact symbolic tests, the 64-state collision enumeration, exact rational grid sweeps, the OU counterexample, a reduced-spectrum limitation witness, and a self-monitoring numerical layer. The monitor can replace floating-point arithmetic by arbitrary precision when overflow, underflow or coefficient coalescence is detected, but it never modifies physical coefficients or projects failed points onto the circle.

References

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